

CHESS MASTERY LIBRARY | 1.d4 + c4 REPERTOIRE

Structures, Model Plans and Training Workbook

The middlegames your 1.d4 repertoire is designed to reach

THE PROMISE

Turn opening recall into playable understanding by learning the recurring pawn skeletons, breaks, piece routes, and warning signs.

COURSE VOLUME 10 OF 10

SEE THE POSITION | RECALL THE PLAN | CALCULATE THE MOVE

What this volume should change

- Identify the structure before choosing a plan.
- Know the characteristic pawn break and best piece route in every major repertoire family.
- Separate a static advantage from the activity required to make it real.
- Use position tests and recall cards to make the entire course retrievable.

STUDY METHOD

Learn each chapter in three layers: first the trigger and plan, then the short main line, then Black's branches. Hide the page and reconstruct the line on a board. Do not memorize notation without the position's purpose.

Inside this volume

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02	Slav and Semi-Slav centers
03	Isolated queen pawn
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Original instructional synthesis. Move sequences were legality-checked with python-chess and critical choices were reviewed with Stockfish 18. Opening-frequency context was checked against the Lichess opening explorer in September 2026. Engines verify tactics; the explanations and training design are the course's teaching layer. Diagrams use Colin M. L. Burnett's Cburnett piece artwork as distributed by Lichess under GPLv2+.

CHAPTER 01

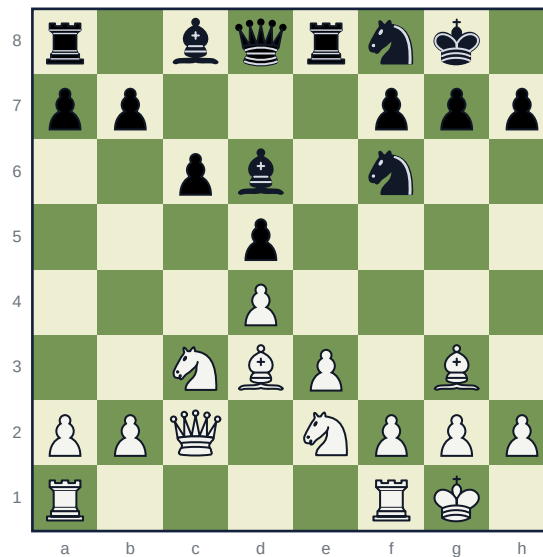
Carlsbad structure

POSITION TRIGGER

White has pawns d4/e3 against Black d5/c6 after cxd5 exd5.

Your anchor line

1. d4 d5 2. c4 e6 3. Nc3 Nf6 4. cxd5 exd5 5. Bf4 c6 6. e3 Bd6 7. Bg3 O-O 8. Bd3 Re8 9. Nge2 Nbd7 10. Qc2 Nf8 11. O-O



Repertoire position after the main line

Why this setup works

The Carlsbad gives White two main plans: minority attack with b4-b5, and central expansion with f3-e4. The choice depends on Black's ...c5 readiness, piece placement, and king safety. Black wants ...c5 or ...e5, piece exchanges, and active outposts. The structure rewards patience because pawn targets often appear only after one side commits a break.

White's plan

- Minority attack when c6 is fixed and the center is secure.
- Central f3-e4 when pieces support the break and ...c5 is controlled.
- Use Ne2-g3/f4, Rab1, and Qc2 as plan-specific routes, not automatic moves.

What Black is trying to do

- ...c5 liquidates c6 and attacks d4.
- ...Ne4 centralizes a piece and supports kingside play.
- ...a5 or ...b5 can stop the minority attack or change its targets.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Fixed c6	Rab1-a3-b4-b5; create a target and entry file.
Passive center	f3-e4 with Qc2, Nge2, and Re1.
Early ...c5	Reclassify as IQP/hanging-pawn play.
...Ne4	Challenge the outpost before launching a wing plan.

MOVE-ORDER DISCIPLINE

The minority attack is not a memorized pawn march. It is a method for creating a weakness when Black cannot liquidate in the center.

MEMORY HOOK

Carlsbad asks one question: target c6 with b5, or open the center with e4?

CHAPTER 02

Slav and Semi-Slav centers

POSITION TRIGGER

Black's c6 pawn supports d5 and the c8 bishop is either outside or inside the chain.

Your anchor line

1. d4 d5 2. c4 c6 3. Nf3 Nf6 4. e3 e6 5. b3 Nbd7 6. Bb2 Bd6 7. Bd3 O-O 8. O-O b6 9. Nbd2 Bb7 10. Qc2



Repertoire position after the main line

Why this setup works

In the Slow Slav, the active bishop on f5 is White's first target; in the Semi-Slav, the trapped bishop makes central breaks ...e5 and ...c5 Black's priority. White uses b3/Bb2 and Nbd2/Qc2 to prepare e4. The structure can change suddenly, so every plan includes a calculation of ...dxc4, ...e5, and ...c5.

White's plan

- Challenge Bf5 with Nh4 in the Slow Slav.
- Build b3-Bb2-Nbd2-Qc2 in the Anti-Meran.
- Prepare e4, but open only when d4 and h2 are tactically secure.

What Black is trying to do

- ...e5 frees the c8 bishop and attacks the center.
- ...c5 attacks d4 and can isolate a pawn.
- ...dxc4 plus ...b5 gains queenside space and tempi on the bishop.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Bishop on f5	Nc3-Nh4; trade it, then use e4.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Bishop on c8	b3-Bb2 and prepare e4 before Black frees it.
...dxc4	Recover by development, then undermine b5 or open center.
...e5 / ...c5	Resolve the center before continuing a wing plan.

MOVE-ORDER DISCIPLINE

The Slav is solid because Black solves one problem at a time. White's edge comes from making each solution cost a concession or tempo.

MEMORY HOOK

Outside bishop: challenge it. Inside bishop: open your b2 bishop and earn e4.

CHAPTER 03

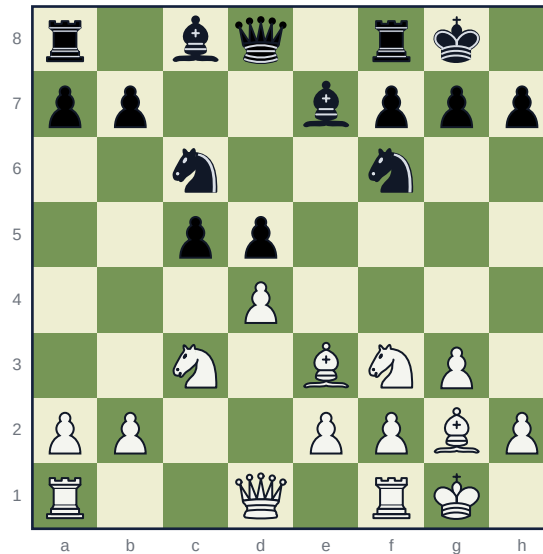
Isolated queen pawn

POSITION TRIGGER

One side has a central d-pawn without a neighboring c- or e-pawn.

Your anchor line

1. d4 d5 2. c4 e6 3. Nc3 c5 4. cxd5 exd5 5. Nf3 Nc6 6. g3 Nf6 7. Bg2 Be7 8. O-O O-O 9. Be3



Repertoire position after the main line

Why this setup works

Against an isolated pawn, blockade first and attack later. The pawn grants its owner active squares and open files, so material pressure before development is dangerous. With the IQP, use piece activity and the d5 advance before exchanges remove the initiative. The same structure can favor either side depending on piece activity.

White's plan

- As blockader: control d4/d5, exchange active pieces, then attack the pawn.
- As owner: centralize rooks and queen, occupy e5/c5, and prepare d5.
- Evaluate every exchange by whether it reduces activity or removes a blockader.

What Black is trying to do

- The IQP side can sacrifice or advance d5 to open lines against the king.
- A knight on e4/c4 can make the pawn untouchable tactically.
- Premature blockade can be driven away if unsupported.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
You oppose IQP	Blockade, exchange activity, attack last.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
You own IQP	Keep pieces, use e5/c5, prepare d5.
d5 break	Calculate opened diagonals and king attacks before pawn count.
Queen trade	Usually helps the blockading side, but verify the resulting endgame activity.

MOVE-ORDER DISCIPLINE

A weakness is not weak if attacking it activates the opponent's pieces. Restraint comes before collection.

MEMORY HOOK

Blockade - exchange activity - attack; or activate - occupy - break if you own it.

CHAPTER 04

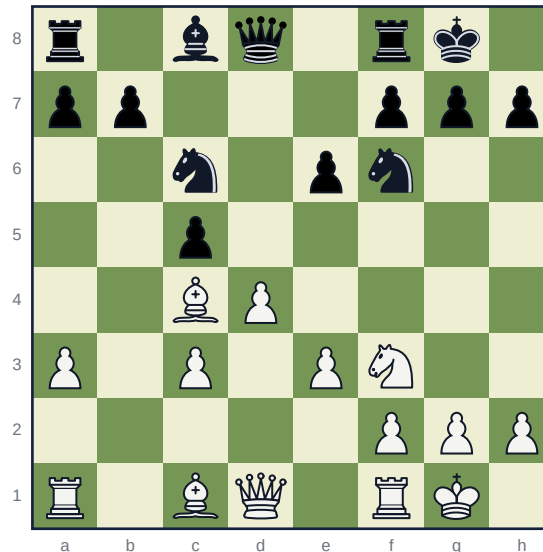
Hanging pawns and mobile centers

POSITION TRIGGER

Connected c- and d-pawns have no neighboring pawns and can advance or become targets.

Your anchor line

1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. e3 O-O 5. Bd3 d5 6. Nf3 c5 7. O-O Nc6 8. a3 Bxc3 9. bxc3 dxc4 10. Bxc4



Repertoire position after the main line

Why this setup works

Hanging pawns control central squares and support active pieces. Their strength is mobility: c5 or d5 can gain space and open lines. Their weakness appears when one advances without support or both are blockaded. The opponent seeks exchanges and pressure; the owner seeks active pieces and a timely break.

White's plan

- Keep pieces active behind the pawns and prepare one controlled advance.
- Use c5 to gain queenside space or d5 to open central lines when supported.
- If opposing them, blockade the square in front and provoke an unsupported advance.

What Black is trying to do

- Pressure on c4/d4 can force the pawns into passive defense.
- Piece exchanges reduce the tactical support behind the pawns.
- An ill-timed advance can leave the other pawn isolated and weak.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Pawns mobile	Improve pieces, then choose c5 or d5 by the opened lines.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
One pawn fixed	Defend actively and seek the other pawn's break.
You oppose them	Blockade, exchange, and provoke.
Endgame near	Avoid unnecessary exchanges if the pawns are your main source of activity.

MOVE-ORDER DISCIPLINE

Connected pawns are not two weaknesses until they are immobilized. Their owner must use them; their opponent must freeze them.

MEMORY HOOK

Mobile means strength; fixed means target.

CHAPTER 05

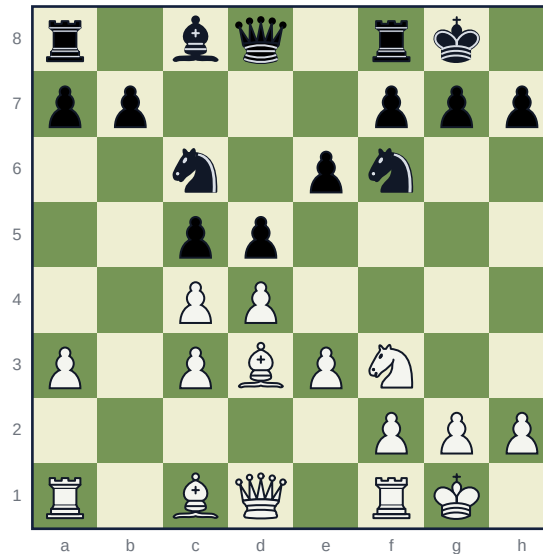
Nimzo doubled c-pawns

POSITION TRIGGER

White has c4/c3 after ...Bxc3+ and bxc3, usually with the bishop pair.

Your anchor line

1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. e3 O-O 5. Bd3 d5 6. Nf3 c5 7. O-O Nc6 8. a3 Bxc3 9. bxc3



Repertoire position after the main line

Why this setup works

The doubled pawns support d4/e4, control b5/d5, and open the b-file. They become weak if Black blockades c4 and attacks c3. White uses the bishop pair by opening the center and moving one c-pawn before the blockade sets. A static endgame may favor Black; an active middlegame often favors White's space and bishops.

White's plan

- Prepare e4 or cxd5/c5 before Black fixes c4.
- Use Rb1 and the bishop pair on open diagonals.
- Avoid mass exchanges that leave both c-pawns fixed without active compensation.

What Black is trying to do

- ...dxc4 and ...cxd4 can fix targets and simplify.
- ...b6/Ba6 exchanges White's strong light bishop.
- ...Nc6-a5 and ...b6 attack c4 repeatedly.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Center fluid	Use e4 or a c-pawn break before blockade.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Center closed	Prepare f4/d5 to release bishops.
c4 blockaded	Create kingside/central play; do not defend passively.
Black offers exchanges	Keep enough pieces to justify the bishop pair and center.

MOVE-ORDER DISCIPLINE

Judge doubled pawns by mobility, square control, open files, and endgame exposure - never by shape alone.

MEMORY HOOK

Use them while mobile; avoid the fixed, simplified version.

CHAPTER 06

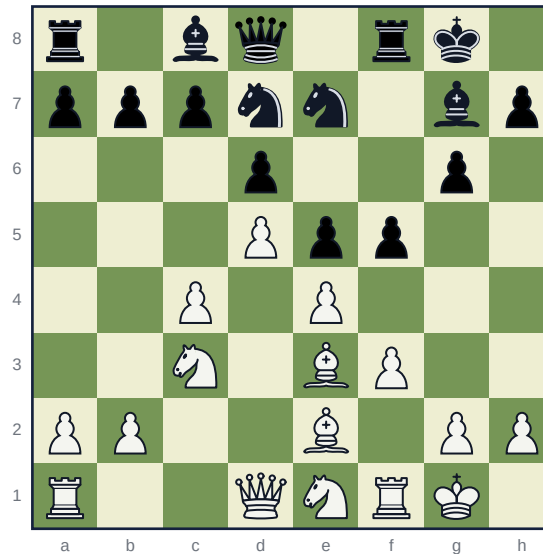
King's Indian closed chains

POSITION TRIGGER

White's pawns d5/e4 face Black d6/e5; the center is locked.

Your anchor line

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7 4. e4 d6 5. Nf3 O-O 6. Be2 e5 7. O-O Nc6 8. d5 Ne7 9. Ne1 Nd7 10. Be3 f5 11. f3



Repertoire position after the main line

Why this setup works

Pawn chains point toward the wing where each side has more space. White attacks queenside with c5/b4; Black attacks kingside with ...f5/g5. The center's closure gives time but not immunity. Every move should either accelerate your break, improve a piece toward the break, or meet a forcing threat.

White's plan

- Route Ne1-d3, use Rc1, and prepare c5/b4.
- Use f3 to secure e4 and blunt the g7 bishop.
- Calculate Black's forcing kingside moves before counting queenside gains.

What Black is trying to do

- ...f5 and ...f4 lock more space and open the f-file.
- ...g5-g4 dislodges defenders and creates mating lines.
- ...Nf4/Ng6 and queen entry near h4 produce sacrifices.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
No immediate threat	Rc1, Nd3, c5 - accelerate counterplay.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
...g4 coming	Choose a concrete defensive move only if it changes the tactic.
c5 succeeds	Use Nb5/d6 and opened c-file entry squares.
Black sacrifices	Calculate checks, captures, and threats before material.

MOVE-ORDER DISCIPLINE

A closed center changes the route of attack, not the requirement to calculate. Wing races are tactical positions wearing strategic clothing.

MEMORY HOOK

Chain points to the wing; tempi and forcing moves decide the race.

CHAPTER 07

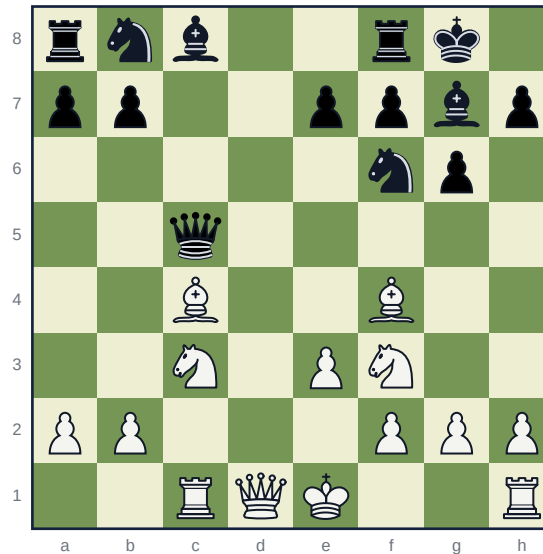
Gruenfeld dynamic center

POSITION TRIGGER

White has a broad center while Black's g7 bishop and ...c5 attack it immediately.

Your anchor line

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. Nf3 Bg7 5. Bf4 O-O 6. e3 c5 7. dxc5 Qa5 8. Rc1 dxc4 9. Bxc4 Qxc5



Repertoire position after the main line

Why this setup works

In the Gruenfeld, center space is valuable only when pieces defend it. Black's piece pressure often forces exchanges, producing open files where development matters more than pawn count. White should return a pawn to complete development when necessary. The bishop on f4 and rook on c1 are valuable because they are active as the center dissolves.

White's plan

- Defend the center with pieces, not additional pawn moves.
- Exchange or return a pawn when it wins development and queen tempi.
- Use open c/d files quickly; castle before trying to punish the queen.

What Black is trying to do

- ...c5 and ...dxc4 remove pawn supports.
- ...Qa5/Qxc5 combines recovery with tactical pressure.
- The g7 bishop becomes strongest when d4/c3 move or disappear.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Center stable	Be2, O-O, Qe2/Rd1; add defenders.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Center attacked twice	Exchange before a pawn becomes a tactical liability.
Queen active	Gain tempi only with developing moves.
Open files	Activity first; a small pawn edge can wait.

MOVE-ORDER DISCIPLINE

A big center is not an achievement until it survives contact with ...c5 and the g7 bishop.

MEMORY HOOK

Pieces hold the center; when it dissolves, use the files first.

CHAPTER 08

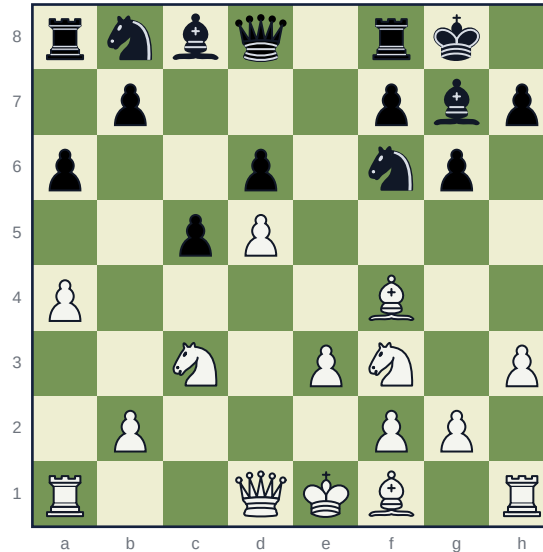
Benoni space wedge

POSITION TRIGGER

White has d5/e4 against Black d6 and a missing e-pawn; ...b5 and ...f5 are Black's breaks.

Your anchor line

1. d4 Nf6 2. c4 c5 3. d5 e6 4. Nc3 exd5 5. cxd5 d6 6. Nf3 g6 7. Bf4 a6 8. a4 Bg7 9. h3 O-O 10. e3



Repertoire position after the main line

Why this setup works

White's space restricts Black but creates a fixed d5 target. Bf4/a4/h3 make Black's freeing breaks harder and preserve key pieces. White often reroutes Nd2-c4 and prepares e5. Black's activity is the price White pays for space, so preventing one break is not enough if the other arrives freely.

White's plan

- Restrain ...b5 with a4 and pressure d6 with Bf4/Nc4.
- Prepare e5 with development and open-file support.
- Track both ...b5 and ...f5; a queenside clamp can be answered in the center/kingside.

What Black is trying to do

- ...b5 opens a/b files and activates rooks.
- ...f5 attacks e4 and opens the g7 bishop/f-file.
- ...Ne5 blockades and exchanges White's active pieces.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
...b5 prepared	Nd2-c4, a5, or axb5 by the resulting files.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
...f5 prepared	Re1 and compare exf5/e5.
...Ne5	Exchange only if the passed d-pawn remains strong.
Black passive	Improve every piece before committing e5.

MOVE-ORDER DISCIPLINE

Space gives maneuvering room, not a permanent advantage. Convert it with a break before Black achieves both counterbreaks.

MEMORY HOOK

Restrain one break, monitor the other, prepare e5 with pieces.

CHAPTER 09

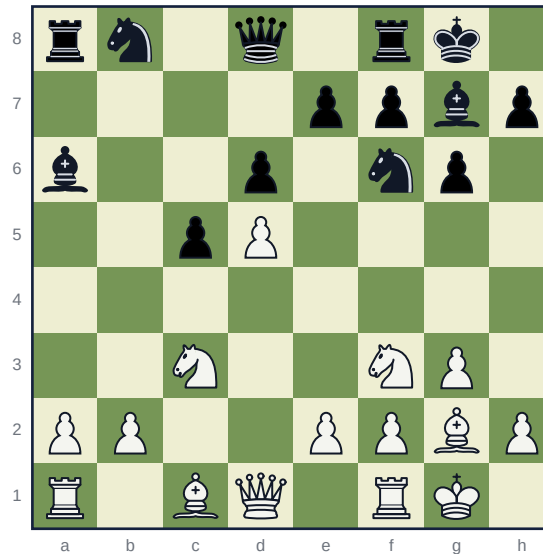
Benko compensation

POSITION TRIGGER

White owns an extra queenside pawn; Black owns open a/b files and active diagonals.

Your anchor line

1. d4 Nf6 2. c4 c5 3. d5 b5 4. cxb5 a6 5. bxa6 Bxa6 6. Nc3 d6 7. Nf3 g6 8. g3 Bg7 9. Bg2 O-O 10. O-O



Repertoire position after the main line

Why this setup works

The material count favors White, but Black's compensation is durable while rooks, queen, and bishops remain active. White neutralizes in layers: contest the dark diagonal, secure b2, control c5/b4, and exchange a major attacking piece. Only then does the extra pawn become an endgame asset. Returning a pawn to complete neutralization is often excellent.

White's plan

- Fianchetto and castle; defend b2 with pieces, not a chain of passive pawns.
- Challenge active rooks and the g7 bishop; exchange compensation.
- Use Nd2-c4 and e4/e5 only after queenside files are controlled.

What Black is trying to do

- Rooks double on a/b files and invade b2.
- Queen checks on a5/c3 and bishop pressure create tactical entry.
- ...e6 attacks d5 and opens central lines while White defends the queenside.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Black doubles rooks	Contest one file or trade one rook; do not defend both passively.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
...e6	Resolve d5 while keeping b2 secure.
Pawn return offered	Return it if the exchange removes files, bishop, or queen pressure.
Major pieces traded	Only then activate the extra pawn and seek conversion.

MOVE-ORDER DISCIPLINE

Material is not usable while the opponent controls the position. Convert activity first, pawn later.

MEMORY HOOK

Neutralize diagonal, b2, files, active pieces - then count the pawn.

CHAPTER 10

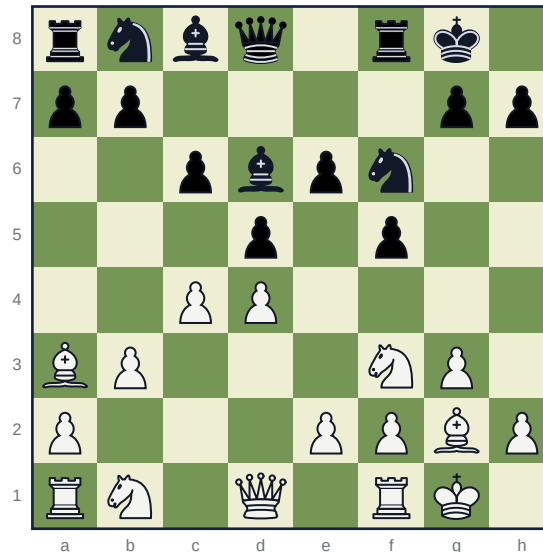
Dutch dark-square strategy

POSITION TRIGGER

Black's f-pawn controls e4 but weakens e6 and the king; the exact pawn triangle selects the target.

Your anchor line

1. d4 f5 2. c4 Nf6 3. g3 e6 4. Bg2 d5 5. Nf3 c6 6. O-O Bd6 7. b3 O-O 8. Ba3



Repertoire position after the main line

Why this setup works

The fianchetto shell gives White safe development and pressure on the center. Against Stonewall, attack e5/c5 dark squares and exchange Bd6. Against Leningrad, interfere with ...e5 using d5. Against Classical setups, prepare e4 with Re1. The common logic is to challenge Black's central claim before attacking the f-pawn.

White's plan

- Use b3/Ba3 and Ne5 against Stonewall.
- Use d5 and long-diagonal pressure against Leningrad.
- Use Re1/e4 against Classical setups before Black achieves ...e5.

What Black is trying to do

- ...e5 frees Black and opens kingside lines.
- ...Ne4 creates a protected attacking outpost.
- Queen routes to e8/h5 create direct threats that outrank positional play.

Branch map

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Stonewall	Exchange Bd6, occupy e5, then prepare a break.

BLACK CHOOSES	YOUR RESPONSE AND PURPOSE
Leningrad	Interfere with ...e5; watch Qe8-h5.
Classical	Re1-e4 while controlling checks.
Unclear setup	Complete fianchetto shell; classify by ...d5/d6/g6.

MOVE-ORDER DISCIPLINE

The f5 pawn is a hook only after Black's center and pieces are controlled. Attack the position's squares first.

MEMORY HOOK

Classify the pawn triangle, then choose e5-square, d5-wedge, or e4-break plan.

Repertoire Test

For every position, name the opening branch, state your strategic objective, and choose a move. The answer is not only a move: it is a move attached to a reason.

TEST 01 | Carlsbad choice

Name White's two principal plans and the cue that selects between them.



Position reached by: 1. d4 d5 2. c4 e6 3. Nc3 Nf6 4. cxd5 exd5 5. Bf4 c6 6. e3 Bd6 7. Bg3 O-O 8. Bd3 Re8 9. Nge2 Nbd7 10. Qc2 Nf8 11. O-O

WRITE BEFORE CHECKING

1) the cue, 2) at least two candidates, 3) Black's strongest reply, and 4) the plan you expect after the move.

TEST 02 | Dynamic weakness

Why should White not attack d5 immediately?



Position reached by: 1. d4 d5 2. c4 e6 3. Nc3 c5 4. cxd5 exd5 5. Nf3 Nc6 6. g3 Nf6 7. Bg2 Be7 8. O-O O-O

WRITE BEFORE CHECKING

1) the cue, 2) at least two candidates, 3) Black's strongest reply, and 4) the plan you expect after the move.

TEST 03 | Nimzo mobility

What makes the doubled c-pawns useful now, and what makes them weak later?



Position reached by: 1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. e3 O-O 5. Bd3 d5 6. Nf3 c5 7. O-O Nc6 8. a3 Bxc3 9. bxc3

WRITE BEFORE CHECKING

1) the cue, 2) at least two candidates, 3) Black's strongest reply, and 4) the plan you expect after the move.

TEST 04 | Closed-center race

Where does each side play, and what is the practical calculation rule?



Position reached by: 1. d4 Nf6 2. c4 g6 3. Nc3 Bg7 4. e4 d6 5. Nf3 O-O 6. Be2 e5 7. O-O Nc6 8. d5 Ne7 9. Ne1 Nd7 10. Be3 f5 11. f3

WRITE BEFORE CHECKING

1) the cue, 2) at least two candidates, 3) Black's strongest reply, and 4) the plan you expect after the move.

TEST 05 | Benko conversion

List the neutralization order before White tries to use the extra pawn.



Position reached by: 1. d4 Nf6 2. c4 c5 3. d5 b5 4. cxb5 a6 5. bxa6 Bxa6 6. Nc3 d6 7. Nf3 g6 8. g3 Bg7 9. Bg2 O-O 10. O-O

WRITE BEFORE CHECKING

1) the cue, 2) at least two candidates, 3) Black's strongest reply, and 4) the plan you expect after the move.

Repertoire Test - Answers

TEST 01 | Carlsbad choice

The plans are the minority attack b4-b5 and the central f3-e4 break. Use the minority attack when c6 is fixed and ...c5 is controlled; use e4 when pieces support it and central opening favors White.

TEST 02 | Dynamic weakness

The isolated pawn supports active pieces and may advance d4. First restrain the break and exchange active pieces. The pawn becomes a genuine target after its dynamic resources are reduced.

TEST 03 | Nimzo mobility

Now they support the center, control key squares, open the b-file, and remain mobile. Later they are weak if Black fixes c4, blockades them, exchanges pieces, and attacks c3.

TEST 04 | Closed-center race

White plays c5/b4 on the queenside; Black plays ...f4/g5/g4 on the kingside. Before every queenside move, calculate Black's forcing checks, captures, and threats. The race is decided by tempi.

TEST 05 | Benko conversion

Contest the long diagonal, secure b2, control b4/c5, and exchange one active rook or bishop. Only after Black's compensation is reduced should White mobilize the extra pawn.

One-Page Recall Card

Cover the right column. Read the trigger and say the memory hook. Then cover the left column and reconstruct which position produces that hook.

White has pawns d4/e3 against Black d5/c6 after cxd5 exd5.	Carlsbad asks one question: target c6 with b5, or open the center with e4?
Black's c6 pawn supports d5 and the c8 bishop is either outside or inside the chain.	Outside bishop: challenge it. Inside bishop: open your b2 bishop and earn e4.
One side has a central d-pawn without a neighboring c- or e-pawn.	Blockade - exchange activity - attack; or activate - occupy - break if you own it.
Connected c- and d-pawns have no neighboring pawns and can advance or become targets.	Mobile means strength; fixed means target.
White has c4/c3 after ...Bxc3+ and bxc3, usually with the bishop pair.	Use them while mobile; avoid the fixed, simplified version.
White's pawns d5/e4 face Black d6/e5; the center is locked.	Chain points to the wing; tempi and forcing moves decide the race.
White has a broad center while Black's g7 bishop and ...c5 attack it immediately.	Pieces hold the center; when it dissolves, use the files first.
White has d5/e4 against Black d6 and a missing e-pawn; ...b5 and ...f5 are Black's breaks.	Restrain one break, monitor the other, prepare e5 with pieces.
White owns an extra queenside pawn; Black owns open a/b files and active diagonals.	Neutralize diagonal, b2, files, active pieces - then count the pawn.
Black's f-pawn controls e4 but weakens e6 and the king; the exact pawn triangle selects the target.	Classify the pawn triangle, then choose e5-square, d5-wedge, or e4-break plan.

SPACED REVIEW

Review today, tomorrow, after three days, after one week, and after one month. A chapter graduates only when you can reconstruct its anchor line and explain its plan without looking.

GAME TRANSFER

After every serious game, locate the first position where your memory ended. Study from that exact fork. This turns the repertoire into a living system while keeping the PDFs as the stable foundation.